

Module 11: Optimal and LRU Page Replacement

1. Optimal (OPT / MIN) vs LRU

Parameter	Optimal (OPT)	Least Recently Used (LRU)
Direction	Looks Forward into future	Looks Backward into past
Fault Rate	Minimum possible benchmark	Very close to Optimal
Belady Anomaly	Never occurs (Stack algo)	Never occurs (Stack algo)
Feasibility	Unrealizable in real-time OS	Implementable via Counter or Stack

Optimal (OPT) vs Least Recently Used (LRU)

String: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1 (Frames = 3)

Optimal (OPT / MIN) Algorithm

Looks FORWARD into the future

- Rule: Us page ko replace karo jo future me **longest period of time tak use nahi hoga.**
- Minimum possible page fault rate guarantee karta hai.
- Practicality: Cannot be implemented in general OS (Requires future oracle knowledge).
- Benchmark: Dusre algorithms ki efficiency measure karne ke liye theoretical baseline ka kaam karta hai.

Total Faults on String = 9 Page Faults

Least Recently Used (LRU) Algorithm

Looks BACKWARD into the past

- Rule: Us page ko replace karo jo past me **longest period of time se access nahi hua hai.**
- OPT ka best practical approximation hai.
- Stack Algorithm: Belady's Anomaly se 100% immune!
- Implementation Methods:
 1. Counter (Logical Clock) - Search min counter on fault
 2. Doubly Linked Stack - Most recent to top, bottom = victim

Total Faults on String = 12 Page Faults

2. 20-Reference Trace Result (3 Frames)

- **Optimal Faults: 9 Faults** (Hit Ratio = 55%)
- **LRU Faults: 12 Faults** (Hit Ratio = 40%)
- **FIFO Faults:** 15 Faults (Suffers from lack of locality).